

PtyGenography: generative priors as regularizers for the phase retrieval problem

(joint work with Selin Aslan, Wim Coene, Tristan van Leeuwen, and Allard Mosk)

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Motivation: ptychography

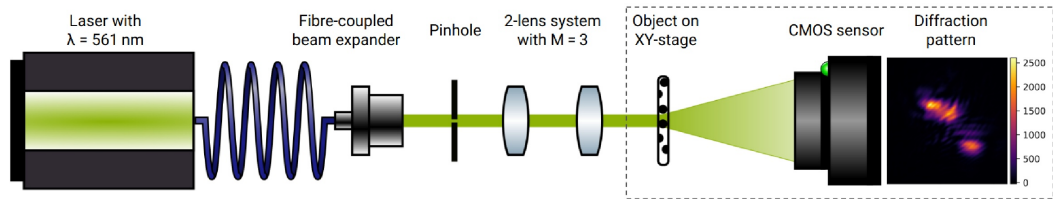


Figure: *Ptychography measurement setup.* The object is moved laterally through the beam, and a CMOS camera sensor records the diffraction intensities. Figure courtesy: J. Seifert, Y. Shao, A.P. Mosk, 2024.

Measurement map: $\mathcal{A} : f \mapsto \{|\mathcal{F}(f \odot p_k)(\ell)|^2\}_{\substack{k \in K \\ \ell \in \Omega}}$, where

- f is the unknown object;
- $p_k = T_k p$ is the probe p shifted to position k .

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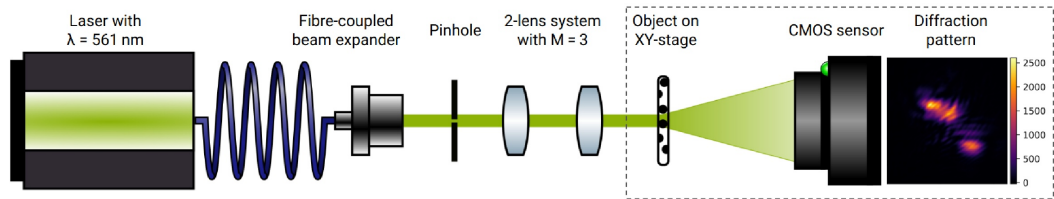


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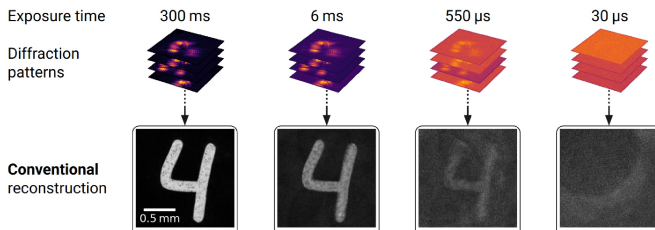
Noisy measurements: $y = \mathcal{A}(f_0) + \varepsilon$, where ε is additive detector noise

Conventional reconstruction approach:

$$\tilde{f} = \arg \min_f \|\mathcal{A}(f) - y\|_2^2 \quad (\text{Conventional})$$

Low signal-to-noise ratio regime

Problem: (Conventional) is **unstable** under low signal-to-noise ratio (SNR)!



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Idea: use **prior information** on the object to **regularize** the inverse problem.

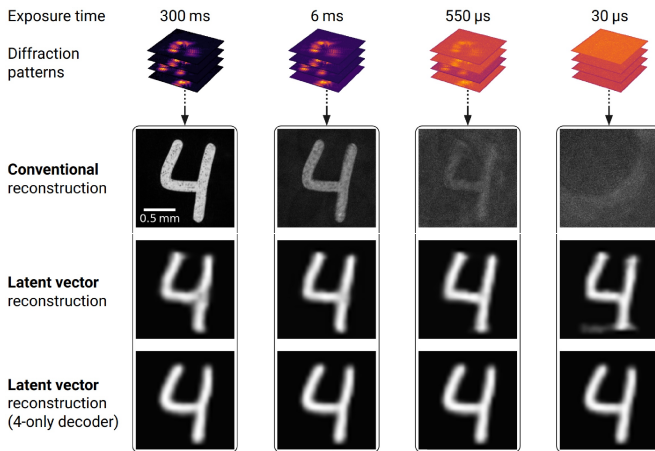


Figure: Performance of conventional vs. generative reconstruction approaches under various SNR values. The object is a handwritten digit '4'. Figure courtesy: J. Seifert, Y. Shao, A.P. Mosk, 2024.

Conventional vs. generative reconstruction

Conventional reconstruction model:

- accurate reconstruction in high SNR
- fails to reconstruct an object in low SNR

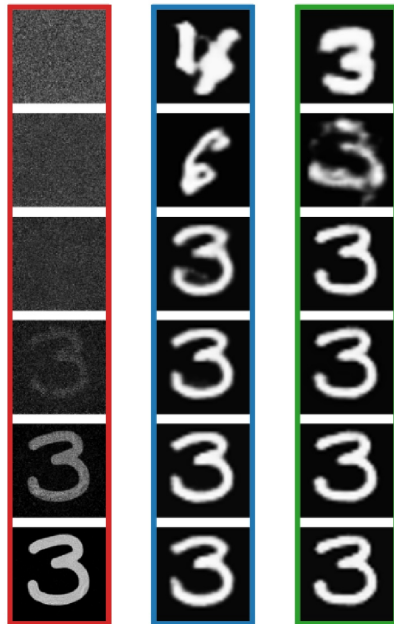
More expressive generative reconstruction (all digits):

- tolerates lower SNR compared to conventional
- can “hallucinate” in very low SNR
- not as accurate as conventional in high SNR

Less expressive generative reconstruction ('3's only):

- object reconstruction even in very low SNR
- fails on out-of-the-distribution objects (with defects)

Question: *Can we take best from both approaches?*



Let us make it formal: mathematical model

Measurement map:

- $\mathcal{A} : [f] \mapsto |A(f)|$, where $[f]$ is an equivalence class for $f \sim e^{i\theta}g$, $\theta \in [0, 2\pi)$; A is a linear map
- general $\mathcal{A}$

Measurements: $y = \mathcal{A}(f_0) + \varepsilon$, where f_0 is the ground truth and ε is noise

Generative model: $f_0 = \mathcal{G}(z_0) + \eta$, where $\mathcal{G}$ is the generative model and η is the object's 'defect'.

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Assumptions:

- both $\mathcal{A}$ and $\mathcal{G}$ are injective and bi-Lipschitz with constants $\alpha, \beta \geq 1$

$$\alpha^{-1} \min_{\theta \in [0, 2\pi)} \|\mathbf{f} - e^{i\theta} \mathbf{f}'\|_2 \leq \|\mathcal{A}(\mathbf{f}) - \mathcal{A}(\mathbf{f}')\|_2 \leq \alpha \min_{\theta \in [0, 2\pi)} \|\mathbf{f} - e^{i\theta} \mathbf{f}'\|_2;$$

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Note: in finite-dimensional setup, the phaseless measurement map $\mathcal{A}(f) = |A(f)|$ satisfies the (bi-)Lipschitz property whenever it is injective (R. Balan, D. Zou, 2016).

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- $\mathcal{G}$ is well-conditioned, in the sense that **$0 < \beta - 1 \ll 1$**
- the **bi-Lipschitz constant γ of $\mathcal{A} \circ \mathcal{G}$** is more favorable than that of $\mathcal{A}$, that is, **$\gamma < \alpha$** .

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Reconstruction approaches:

$$\tilde{f} = \arg \min_f \|\mathcal{A}(f) - y\|_2^2$$

(Conventional)

$$\begin{aligned}\tilde{z} &= \arg \min_z \|\mathcal{A} \circ \mathcal{G}(z) - y\|_2^2 \\ \tilde{f} &= \mathcal{G}(\tilde{z})\end{aligned}$$

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- evaluate the **reconstruction error** of (Conventional) and (Generative);
- propose a practical way to **detect the bias** introduced by the generative model;
- propose a **unified reconstruction method** that bridges (Conventional) and (Generative) and performs well for different noise levels.

Reconstruction error

Lemma (Conventional reconstruction)

Let $\tilde{f} = \arg \min_f \|\mathcal{A}(f) - y\|_2$ with $y = \mathcal{A}(f_0) + \varepsilon$. Then the reconstruction error is given by

$$\|\tilde{f} - f_0\|_2 \leq 2\alpha\|\varepsilon\|_2.$$

Lemma (Generative reconstruction)

Suppose $f_0 = \mathcal{G}(z_0) + \eta$ and $y = \mathcal{A}(f_0) + \varepsilon$. Let $\tilde{z} = \arg \min_z \|\mathcal{A} \circ \mathcal{G}(z) - y\|_2$ and $\tilde{f} = \mathcal{G}(\tilde{z})$. Then the reconstruction error is bounded by

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Out-of-distribution case ($\eta \neq 0$):

- less noise amplification at the cost of introducing a bias term
- a trade-off between the introduced bias and the noise amplification

Detecting bias: Vanderlugt correlation

Question: given a solution $\tilde{z} = \arg \min_z \|\mathcal{A} \circ \mathcal{G}(z) - y\|_2^2$ to (Generative), can we detect/evaluate the bias $\|\mathcal{G}(\tilde{z}) - f_0\|$?

Answer depends on the structural properties of $\mathcal{A}$ and $\mathcal{G}$...

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$$\mathcal{F}(\eta)(j) \approx \frac{y(j) - |\mathcal{F}(f)(j)|^2}{(\mathcal{F}(f)(j))^*} - \frac{(\mathcal{F}(\eta)(j))^* \mathcal{F}(f)(j)}{(\mathcal{F}(f)(j))^*}$$

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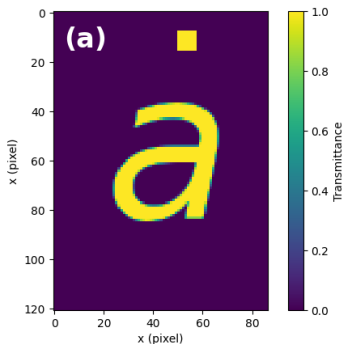
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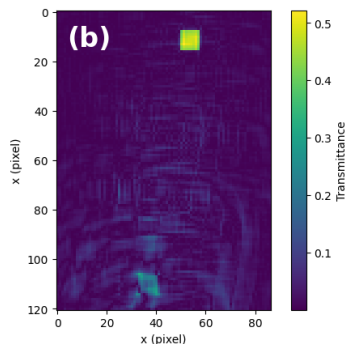
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Vanderlugt correlation: example

- 'Perfect' object $\mathcal{G}(z_0)$: letter 'a' without a dot accent
- 'Imperfection' η : the dot accent



Left: ground truth object $f_0 = \mathcal{G}(z_0) + \eta$



Right: reconstructed approximation of η

$$(\tilde{z}, \tilde{f}) = \arg \min_{z, f} \|\mathcal{A}(f) - y\|_2^2 + \lambda^2 \|\mathcal{G}(z) - f\|_2^2$$

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Lemma (Unified reconstruction)

The estimate $\tilde{f}$ resulting from (Unified) has a reconstruction error bounded as

$$\|\tilde{f} - f_0\|_2 \leq \lambda \alpha \|\eta\|_2 + 2\alpha \|\varepsilon\|_2.$$

Note: The regularization parameter λ should be adaptively selected based on the noise intensity $\|\varepsilon\|_2$. Letting $\lambda \propto \|\varepsilon\|_2$, the reconstruction error of (Unified) is bounded as

$$\|\tilde{f} - f_0\|_2 \lesssim \|\varepsilon\|_2.$$

Numerics: experiment setup

Measurement map: $n = 64$, $\mathcal{A}: \mathbb{C}^n \rightarrow \mathbb{R}^m$, $\mathcal{A}(f) = |Af|^2$



- masked Fourier transform $A = \begin{pmatrix} F \text{diag}(a_1) \\ \vdots \\ F \text{diag}(a_\ell) \end{pmatrix}$
- $F \in \mathbb{C}^{n \times n}$ is the discrete Fourier transform matrix
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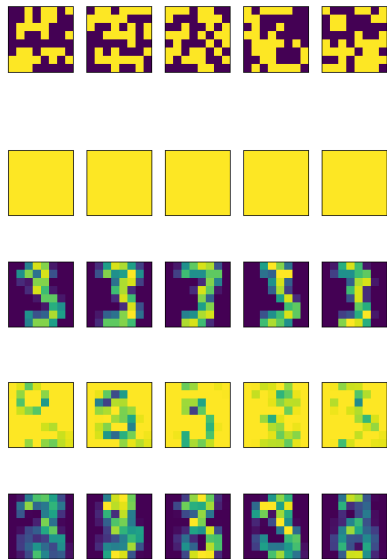
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Generative model: $k = 30$, $\mathcal{G}: \mathbb{C}^k \rightarrow \mathbb{C}^n$, $\mathcal{G}(z) = Gz + b$

- “model” data set: handwritten digits
- $b \in \mathbb{C}^n$ and $G \in \mathbb{R}^{n \times k}$ are obtained by principle component analysis



Numerical results

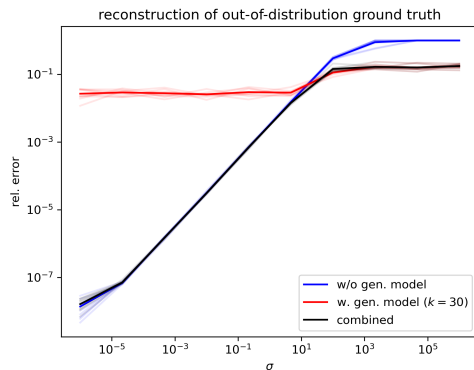
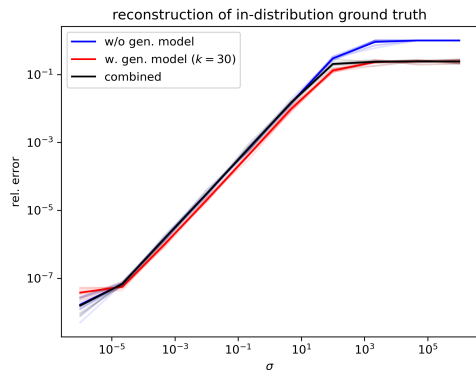


Figure: Reconstruction errors for (Conventional), (Generative), and (Unified), as a function of the noise variance σ . **Left:** in-distribution ground truth ($\eta = 0$); **Right:** out-of-distribution ground truth ($\eta \neq 0$).

- Generative priors can improve the robustness of the inverse problem solution to measurement noise, at the expense of introducing a bias in the reconstruction.
- We proposed a method that combines the best characteristics of conventional and generative approaches by interpolating between them.
- The presented error bounds can be improved with more careful analysis. Further research is needed to solidify our understanding of the unified approach, to refine it, and to make it feasible for high-dimensional problems.

Thank You for Your Attention!